# The Conditional Activation Model: Uncovering Policy Demand Responses to AI-Driven Automation

Ignacio Urbina

March 9, 2026

Stony Brook University

# Outline

# Puzzle & Theory

# The Automation–Policy Demand Puzzle

- AI and automation may displace up to 47% of U.S. jobs *(Frey & Osborne, 2017)*.

# The Automation–Policy Demand Puzzle

- AI and automation may displace up to 47% of U.S. jobs *(Frey & Osborne, 2017)*.
- Standard logic: information about **personal risk** $\rightarrow$ higher demand for government protection *(Margalit, 2019)*.

# The Automation–Policy Demand Puzzle

- AI and automation may displace up to 47% of U.S. jobs *(Frey & Osborne, 2017)*.
- Standard logic: information about **personal risk** $\rightarrow$ higher demand for government protection *(Margalit, 2019)*.
- But experiments produce a **mixed pattern**: some find null effects, others conditional or positive effects.

# The Automation–Policy Demand Puzzle

- AI and automation may displace up to 47% of U.S. jobs *(Frey & Osborne, 2017)*.
- Standard logic: information about **personal risk** $\rightarrow$ higher demand for government protection *(Margalit, 2019)*.
- But experiments produce a **mixed pattern**: some find null effects, others conditional or positive effects.

$\Rightarrow$ **When—and for whom—does exposure to automation risk translate into policy demand?**

# Mixed Findings in the Literature

| Study | Trust Modeled? | Prior Beliefs Elicited? | Engagement Required? | Policy Framing | Finding |
|---|---|---|---|---|---|
| Zhang (2022) | No | No | No (vignette) | Mixed | Null |
| Magistro et al. (2024) | No | No | No (conjoint) | Contextualized | Null |
| Ladreit (2022) | No | No | No (vignette) | Contextualized | Null |
| Arntz et al. (2024) | No | Yes (optimism) | No | Traditional | **Heterogeneous** |
| Heinrich & Witko (2024) | Proxy (knowledge) | No | No | Mixed | **Conditional+** |
| Haslberger (2024) | Reduced (behavioral) | No | Yes (AI use) | Traditional | **Positive** |
| **This Study** | **Yes (latent)** | **Yes (Gap)** | **Yes (3-step)** | **Contextualized** | **CATE** |

Studies without trust modeling or belief elicitation $\rightarrow$ null effects.

Design features accumulate $\rightarrow$ conditional effects emerge.

## Gate 1 — Source Trust

- Source credibility shapes whether information is processed at all *(Druckman, 2001)*.

# Theoretical Ingredients

## Gate 1 — Source Trust

- Source credibility shapes whether information is processed at all *(Druckman, 2001)*.
- Trust in scientific expertise is increasingly polarized *(Gauchat, 2012; Pew, 2024)*.

## Gate 1 — Source Trust

- Source credibility shapes whether information is processed at all *(Druckman, 2001)*.
- Trust in scientific expertise is increasingly polarized *(Gauchat, 2012; Pew, 2024)*.

⇒ **Skeptics** reject the expert signal before it can activate anything.

# Theoretical Ingredients

## Gate 1 — Source Trust

- Source credibility shapes whether information is processed at all *(Druckman, 2001)*.
- Trust in scientific expertise is increasingly polarized *(Gauchat, 2012; Pew, 2024)*.

$\Rightarrow$ **Skeptics** reject the expert signal before it can activate anything.

## Gate 2 — Belief Gap $\times$ Dispositional Anxiety

- AIT: **trait anxiety** (a stable disposition, not momentary worry) activates vigilance and accuracy-motivated processing *(Marcus et al., 2000; Valentino et al., 2008)*.

# Theoretical Ingredients

## Gate 1 — Source Trust

- Source credibility shapes whether information is processed at all *(Druckman, 2001)*.
- Trust in scientific expertise is increasingly polarized *(Gauchat, 2012; Pew, 2024)*.

$\Rightarrow$ **Skeptics** reject the expert signal before it can activate anything.

## Gate 2 — Belief Gap $\times$ Dispositional Anxiety

- AIT: **trait anxiety** (a stable disposition, not momentary worry) activates vigilance and accuracy-motivated processing *(Marcus et al., 2000; Valentino et al., 2008)*.
- High-disposition individuals override partisan heuristics and engage effortfully with threatening information *(Albertson & Gadarian, 2015)*.

# Theoretical Ingredients

## Gate 1 — Source Trust

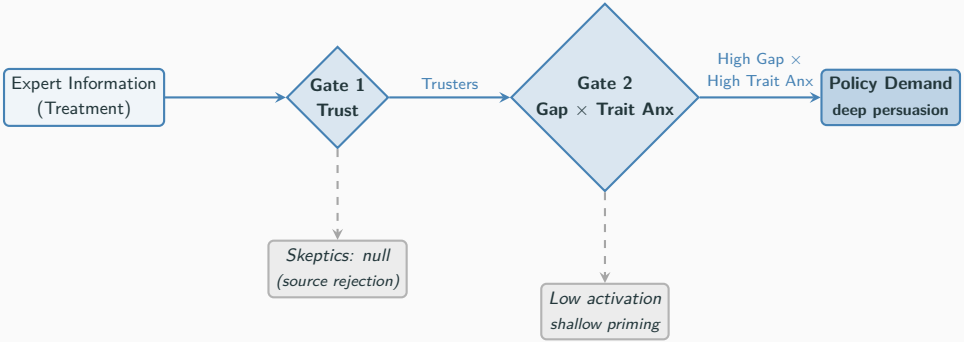
- Source credibility shapes whether information is processed at all *(Druckman, 2001)*.
- Trust in scientific expertise is increasingly polarized *(Gauchat, 2012; Pew, 2024)*.

$\Rightarrow$ **Skeptics** reject the expert signal before it can activate anything.

## Gate 2 — Belief Gap $\times$ Dispositional Anxiety

- AIT: **trait anxiety** (a stable disposition, not momentary worry) activates vigilance and accuracy-motivated processing *(Marcus et al., 2000; Valentino et al., 2008)*.
- High-disposition individuals override partisan heuristics and engage effortfully with threatening information *(Albertson & Gadarian, 2015)*.
- **Belief Gap** = expert estimate − prior belief: the "activation potential" for updating.

# The Conditional Activation Model



# Hypotheses

**H1. Weak Main Effect.** Positive but modest **unconditional ATE**—pooling dilutes the responsive subgroup.

# Hypotheses

**H1. Weak Main Effect.** Positive but modest **unconditional ATE**—pooling dilutes the responsive subgroup.

**H2. Gate 1 Gradient.** $\tau_{\mathrm{T}} > \tau_{\mathrm{S}}$—source rejection blocks information uptake.

# Hypotheses

**H1. Weak Main Effect.** Positive but modest **unconditional ATE**—pooling dilutes the responsive subgroup.

**H2. Gate 1 Gradient.** $\tau_{\mathrm{T}} > \tau_{\mathrm{S}}$—source rejection blocks information uptake.

**H3. Belief Gap Dose.** Among Trusters, effects increase with **positive Belief Gap**.

# Hypotheses

**H1. Weak Main Effect.** Positive but modest **unconditional ATE**—pooling dilutes the responsive subgroup.

**H2. Gate 1 Gradient.** $\tau_{\mathrm{T}} > \tau_{\mathrm{S}}$—source rejection blocks information uptake.

**H3. Belief Gap Dose.** Among Trusters, effects increase with **positive Belief Gap**.

**H4. Anxiety Amplification.** Among Trusters with positive Gaps, **high trait anxiety** amplifies effects.

# Hypotheses

**H1. Weak Main Effect.** Positive but modest **unconditional ATE**—pooling dilutes the responsive subgroup.

**H2. Gate 1 Gradient.** $\tau_{\mathrm{T}} > \tau_{\mathrm{S}}$—source rejection blocks information uptake.

**H3. Belief Gap Dose.** Among Trusters, effects increase with **positive Belief Gap**.

**H4. Anxiety Amplification.** Among Trusters with positive Gaps, **high trait anxiety** amplifies effects.

**H5. Conditional Activation.** Largest effects among Trusters with high anxiety *and* positive Gaps—both gates satisfied.

# The Right Estimand

**Why the ATE understates the story:**

$$\text{ATE} = \underbrace{\pi}_{\text{Truster share}} \cdot \tau_{\mathrm{T}} + \underbrace{(1-\pi)}_{\text{Skeptic share}} \cdot \underbrace{\tau_{\mathrm{S}}}_{\text{flat}}$$

# The Right Estimand

**Why the ATE understates the story:**

$$\text{ATE} = \underbrace{\pi}_{\text{Truster share}} \cdot \tau_{\mathrm{T}} + \underbrace{(1-\pi)}_{\text{Skeptic share}} \cdot \underbrace{\tau_{\mathrm{S}}}_{\text{flat}}$$

If $\tau_{\mathrm{S}}$ is positive but **unmoderated** (no response to Gap or Anxiety), the ATE **dilutes** the conditional signal. The theoretically relevant quantity is the **CATE**.

# Research Design

# Sample and Randomization Balance

## Sample

- $N = 901$ U.S. workers via CloudResearch Connect (Treatment: 475 / Control: 426)
- Targeting (enforced): ages 18–65; full-time, part-time, business owner, or unemployed; English-speaking

# Sample and Randomization Balance

## Sample

- $N = 901$ U.S. workers via CloudResearch Connect (Treatment: 475 / Control: 426)
- Targeting (enforced): ages 18–65; full-time, part-time, business owner, or unemployed; English-speaking

## Balance

- All 13 pre-treatment covariates: $|\text{SMD}| < 0.1$; only trait anxiety nominally significant ($p = .030$), does not survive Holm correction

# Sample and Randomization Balance

## Sample

- $N = 901$ U.S. workers via CloudResearch Connect (Treatment: 475 / Control: 426)
- Targeting (enforced): ages 18–65; full-time, part-time, business owner, or unemployed; English-speaking

## Balance

- All 13 pre-treatment covariates: $|\text{SMD}| < 0.1$; only trait anxiety nominally significant ($p = .030$), does not survive Holm correction

## Manipulation checks

- 93.4% comprehension; perceived gap tracks objective gap ($\rho = 0.73$); belief updating tracks gap ($\rho = 0.66$)

# Treatment: Three-Step Belief Correction

## Step 1: Neutral context

### Automation and Your Job

Artificial intelligence (AI) is getting better at doing work people used to do. Computers already answer customer questions, write basic code, and sort medical images. *As AI improves, some jobs may be partly-or fully-done by machines, while new kinds of jobs also appear.*

### How do experts judge a job's automation risk?

Experts look at the small tasks that make up a job. When those tasks are routine, clearly defined, and follow repeated patterns-and there are lots of examples for computers to learn from-AI systems can pick them up more easily.
The shorter and more repetitive a task is, the easier it is to automate.

| EXAMPLE JOB | TYPICAL TASK | AI RISK (TODAY) |
|---|---|---|
| IT Help Desk Technician | 10-minute help IT tickets that follow a script | *Higher Risk* – chatbots can handle most tickets |
| IT Systems Designer | Plan how the company's tech infrastructure will work over the next year | *Lower Risk* – needs long-term planning and team buy-in |

### Other factors also influence how likely a job is to be replaced by AI automation:

**Jobs are at greater risk when they:**

- Can be done entirely online, or
- Have lots of recorded data for AI to learn from.

**Jobs are less at risk when they:**

- Require in-person skills or dexterity, or
- Rely on human trust or empathy.

## Step 2: Elicit prior belief

**Imagine every worker in the country standing on one long line, sorted by how likely their job is to be automated.**
The very front of the line (position 0) belongs to people whose jobs are safer than almost everyone else's, while the very back (position 100) belongs to those whose jobs are at risk sooner than almost everyone else's.
Someone standing in the middle of the line (position 50) would have half the number of workers ahead of them and half behind.

- **Safest** jobs are those whose chances of being automated are below most other people.
- **Riskier** jobs are those with an automation potential above most other people.

**Taking a moment to reflect on your own job, where do you feel you would place yourself on that line right now?**

0=Safest than everyone else's. Riskier than everyone else's=100.
0 10 20 30 40 50 60 70 80 90 100

*Click and drag on the slider to select your occupation's relative risk of automation.*

## Step 3: Reveal expert & gap

### Comparing Your Job Automation Risk: Your View vs. Expert Insight

Earlier, we asked you to imagine all workers standing in a single line, ranked by the risk of their jobs being automated — from the lowest relative risk (position 0) to the highest relative risk (position 100).

You placed yourself at: **Position 37**

Researchers place your job at around: **Position 82**

This means that, according to experts, your job has a **higher** risk of automation than **82** out of 100 people — but a **lower** risk than **18** others.

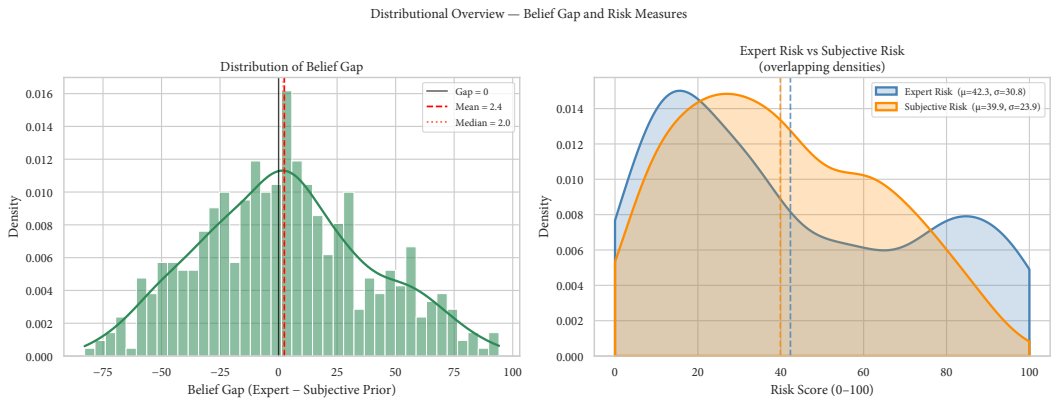
Regarding how likely is your job to be automated in the future relative to others, how does the experts automation risk assessment compare to your initial view?

- ○ The experts place my job at a **significantly higher** relative risk
- ○ The experts place my job at a slightly higher relative risk
- ○ The experts place my job at a fairly similar relative risk
- ○ The experts place my job at a slightly lower relative risk
- ○ The experts place my job at a **significantly lower** relative risk

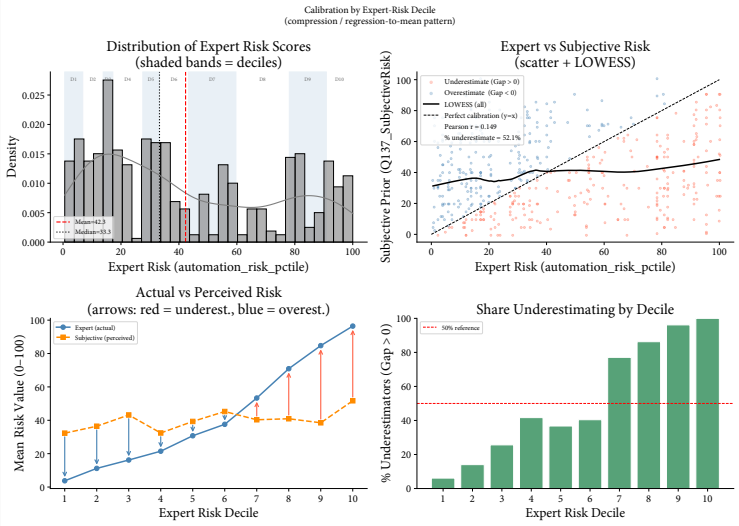
*Control group*: skips all steps; proceeds directly to outcome battery.

**Belief Gap** = Expert estimate − Prior belief (computed, not shown to participants).

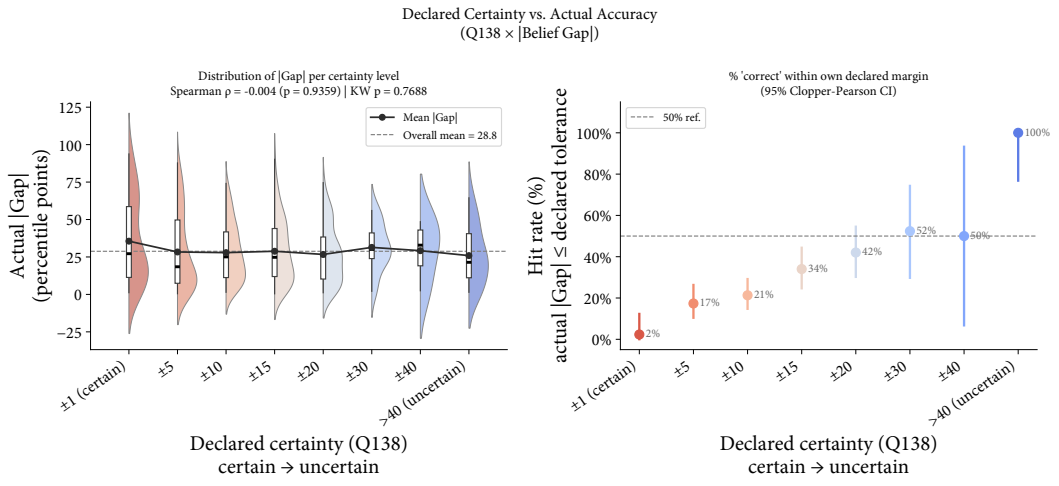
# The Belief Gap: Construction and Distribution



# Belief Gap: Systematic Miscalibration



# Belief Gap: Workers Don't Know How Wrong They Are



# Manipulation Checks

**Perceived Gap Direction (Q141)**

| Category | $n$ | Mean Gap |
|---|---|---|
| Sig. higher risk | 143 | 35.4 |
| Slightly higher risk | 98 | 6.2 |
| Fairly similar risk | 71 | -2.0 |
| Slightly lower risk | 73 | -14.4 |
| Sig. lower risk | 97 | -33.4 |
| Spearman $\rho = -0.733 ***$ | | |

**Belief Updating (Q143)**

| Category | $n$ | Mean Gap |
|---|---|---|
| Much lower than prior | 55 | -35.6 |
| A little lower than prior | 78 | -23.5 |
| About the same as prior | 172 | 3.4 |
| A little higher than prior | 93 | 15.6 |
| Much higher than prior | 84 | 35.6 |
| Spearman $\rho = 0.656 ***$ | | |

Comprehension check accuracy: 93.4% (treated + control).

# Measurement: Scales and Psychometric Properties

## Outcome

- **Policy Demand** (Q96): 5 items, 7-pt scale ($\alpha = .82$, $\omega = .82$)

  Unemployment benefits, AI regulation, retraining, labor protections, AI job limits

# Measurement: Scales and Psychometric Properties

## Outcome

- **Policy Demand** (Q96): 5 items, 7-pt scale ($\alpha = .82$, $\omega = .82$)
  Unemployment benefits, AI regulation, retraining, labor protections, AI job limits

## Intermediate outcomes

- **Automation Anxiety** (Q94): 4 items, 7-pt scale ($\alpha = .93$, $\omega = .92$)
- **Perceived AI Impact** (Q95): 7 items, 5-pt scale; 2-factor (threat/opportunity)

# Measurement: Scales and Psychometric Properties

## Outcome

- **Policy Demand** (Q96): 5 items, 7-pt scale ($\alpha = .82$, $\omega = .82$)
  Unemployment benefits, AI regulation, retraining, labor protections, AI job limits

## Intermediate outcomes

- **Automation Anxiety** (Q94): 4 items, 7-pt scale ($\alpha = .93$, $\omega = .92$)
- **Perceived AI Impact** (Q95): 7 items, 5-pt scale; 2-factor (threat/opportunity)

## Moderators

- **Gate 1 — Trust** (Q142): single item, 5-pt scale (post-treatment, treated only)
- **Gate 2 — Trait Anxiety** (Q23): 4 items, 7-pt scale ($\alpha = .91$, $\omega = .91$)
- **Belief Gap**: Expert percentile − Subjective prior (continuous, 0–100)

# Measurement: Scales and Psychometric Properties

## Outcome

- **Policy Demand** (Q96): 5 items, 7-pt scale ($\alpha = .82$, $\omega = .82$)
  Unemployment benefits, AI regulation, retraining, labor protections, AI job limits

## Intermediate outcomes

- **Automation Anxiety** (Q94): 4 items, 7-pt scale ($\alpha = .93$, $\omega = .92$)
- **Perceived AI Impact** (Q95): 7 items, 5-pt scale; 2-factor (threat/opportunity)

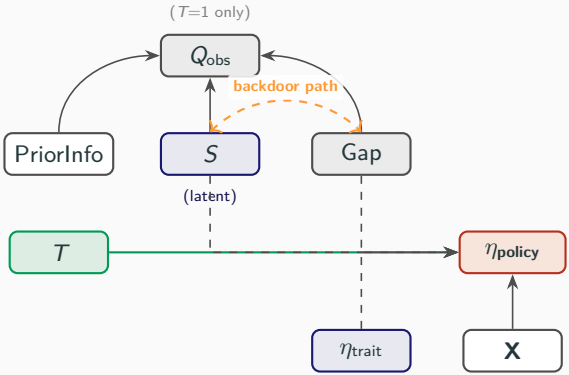
## Moderators

- **Gate 1 — Trust** (Q142): single item, 5-pt scale (post-treatment, treated only)
- **Gate 2 — Trait Anxiety** (Q23): 4 items, 7-pt scale ($\alpha = .91$, $\omega = .91$)
- **Belief Gap**: Expert percentile − Subjective prior (continuous, 0–100)

Joint 6-factor CFA (23 items): CFI = .958, RMSEA = .054, SRMR = .048

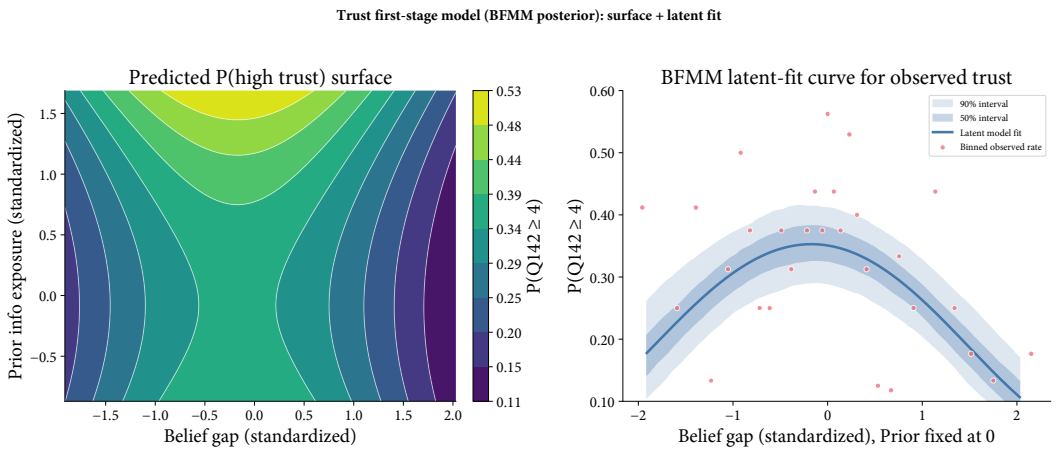
# Identification & Model

# The Identification Problem: Trust as a Collider




$Q_{obs}$ is a collider: conditioning on observed trust opens a spurious path between Gap and $S$. $\Rightarrow$ Cannot simply split on observed trust.

# Trust Covariates: Epistemic Validation



# Bayesian Finite Mixture Model

1. **Latent trust classification** (ordered logit, 2 components)
   - $Q_{\text{obs}} \mid S, \text{Gap}, \text{PriorInfo} \sim \text{OrderedLogit}(\delta S + \gamma' \mathbf{z}, \kappa)$
   - Covariates $\mathbf{z}$: Gap, Gap$^2$, PriorInfo, PriorInfo$^2$ (blocks collider)
   - Constraint: $\delta > 0$ (resolves label switching)

# Bayesian Finite Mixture Model

1. **Latent trust classification** (ordered logit, 2 components)
   - $Q_{\text{obs}} \mid S, \text{Gap}, \text{PriorInfo} \sim \text{OrderedLogit}(\delta S + \gamma' \mathbf{z}, \kappa)$
   - Covariates $\mathbf{z}$: Gap, Gap$^2$, PriorInfo, PriorInfo$^2$ (blocks collider)
   - Constraint: $\delta > 0$ (resolves label switching)
2. **CFA measurement models** (raw 1–7 scales)
   - $\eta_{\text{trait}}$: 4 trait-anxiety items ($\alpha = 0.91$)
   - $\eta_{\text{policy}}$: 5 policy-demand items ($\alpha = 0.82$)

# Bayesian Finite Mixture Model

1. **Latent trust classification** (ordered logit, 2 components)
   - $Q_{\text{obs}} \mid S, \text{Gap}, \text{PriorInfo} \sim \text{OrderedLogit}(\delta S + \gamma' \mathbf{z}, \kappa)$
   - Covariates $\mathbf{z}$: Gap, Gap$^2$, PriorInfo, PriorInfo$^2$ (blocks collider)
   - Constraint: $\delta > 0$ (resolves label switching)
2. **CFA measurement models** (raw 1–7 scales)
   - $\eta_{\text{trait}}$: 4 trait-anxiety items ($\alpha = 0.91$)
   - $\eta_{\text{policy}}$: 5 policy-demand items ($\alpha = 0.82$)
3. **Class-specific structural equations**
   - $\tau_{\text{T}} = \beta_0^T + \beta_1^T \cdot \text{Gap} + \beta_2^T \cdot \eta_{\text{trait}} + \beta_3^T \cdot (\text{Gap} \times \eta_{\text{trait}})$
   - $\tau_{\text{S}} = \beta_0^S + \ldots$ (same form, different parameters)
   - Constraint: $\beta_\Delta \geq 0$ (Trusters $\geq$ Skeptics monotonicity)

# Bayesian Finite Mixture Model

1. **Latent trust classification** (ordered logit, 2 components)
   - $Q_{\text{obs}} \mid S, \text{Gap}, \text{PriorInfo} \sim \text{OrderedLogit}(\delta S + \gamma' \mathbf{z}, \kappa)$
   - Covariates $\mathbf{z}$: Gap, Gap$^2$, PriorInfo, PriorInfo$^2$ (blocks collider)
   - Constraint: $\delta > 0$ (resolves label switching)
2. **CFA measurement models** (raw 1–7 scales)
   - $\eta_{\text{trait}}$: 4 trait-anxiety items ($\alpha = 0.91$)
   - $\eta_{\text{policy}}$: 5 policy-demand items ($\alpha = 0.82$)
3. **Class-specific structural equations**
   - $\tau_{\text{T}} = \beta_0^T + \beta_1^T \cdot \text{Gap} + \beta_2^T \cdot \eta_{\text{trait}} + \beta_3^T \cdot (\text{Gap} \times \eta_{\text{trait}})$
   - $\tau_{\text{S}} = \beta_0^S + \ldots$ (same form, different parameters)
   - Constraint: $\beta_\Delta \geq 0$ (Trusters $\geq$ Skeptics monotonicity)

Estimation: Stan/HMC-NUTS, 4 chains $\times$ 2,000 draws. All $\hat{R} < 1.01$.

# Structural Model (Gate 2)

Latent policy demand decomposes into structural mean + residual:

$$\eta_{\text{policy},i} = \mu_{\text{policy},i} + \sigma_{\text{struct}} \cdot \zeta_i, \quad \zeta_i \sim \mathcal{N}(0,1)$$

# Structural Model (Gate 2)

Latent policy demand decomposes into structural mean + residual:

$$\eta_{\text{policy},i} = \mu_{\text{policy},i} + \sigma_{\text{struct}} \cdot \zeta_i, \quad \zeta_i \sim \mathcal{N}(0,1)$$

The structural mean is a mixture of type-specific components:

$$\mu_{\text{policy},i} = \pi \cdot \mu_{\text{Truster},i} + (1-\pi) \cdot \mu_{\text{Skeptic},i}$$

# Structural Model (Gate 2)

Latent policy demand decomposes into structural mean + residual:

$$\eta_{\text{policy},i} = \mu_{\text{policy},i} + \sigma_{\text{struct}} \cdot \zeta_i, \quad \zeta_i \sim \mathcal{N}(0,1)$$

The structural mean is a mixture of type-specific components:

$$\mu_{\text{policy},i} = \pi \cdot \mu_{\text{Truster},i} + (1-\pi) \cdot \mu_{\text{Skeptic},i}$$

For treated units ($T_i = 1$):

$$\mu_{\text{Truster},i} = (\alpha_S + \alpha_\Delta) + \tau_{\text{T}}(\text{Gap}_i,\ \eta_{\text{trait},i}) + \mathbf{X}_i'\boldsymbol{\beta}_{\text{ctrl}}$$

$$\mu_{\text{Skeptic},i} = \alpha_S + \tau_{\text{S}}(\text{Gap}_i,\ \eta_{\text{trait},i}) + \mathbf{X}_i'\boldsymbol{\beta}_{\text{ctrl}}$$

# Structural Model (Gate 2)

Latent policy demand decomposes into structural mean + residual:

$$\eta_{\text{policy},i} = \mu_{\text{policy},i} + \sigma_{\text{struct}} \cdot \zeta_i, \quad \zeta_i \sim \mathcal{N}(0,1)$$

The structural mean is a mixture of type-specific components:

$$\mu_{\text{policy},i} = \pi \cdot \mu_{\text{Truster},i} + (1-\pi) \cdot \mu_{\text{Skeptic},i}$$

For treated units ($T_i = 1$):

$$\mu_{\text{Truster},i} = (\alpha_S + \alpha_\Delta) + \tau_{\text{T}}(\text{Gap}_i,\ \eta_{\text{trait},i}) + \mathbf{X}_i'\boldsymbol{\beta}_{\text{ctrl}}$$

$$\mu_{\text{Skeptic},i} = \alpha_S + \tau_{\text{S}}(\text{Gap}_i,\ \eta_{\text{trait},i}) + \mathbf{X}_i'\boldsymbol{\beta}_{\text{ctrl}}$$

The CATE for each class:

$$\tau_S(\text{Gap},\ \eta_{\text{trait}}) = \beta_{S,1} + \beta_{S,2} \cdot \text{Gap} + \beta_{S,3} \cdot \eta_{\text{trait}} + \beta_{S,4} \cdot \underbrace{\text{Gap} \times \eta_{\text{trait}}}_{\text{key interaction (H5)}}$$

# Structural Model (Gate 2)

Latent policy demand decomposes into structural mean + residual:

$$\eta_{\text{policy},i} = \mu_{\text{policy},i} + \sigma_{\text{struct}} \cdot \zeta_i, \quad \zeta_i \sim \mathcal{N}(0,1)$$

The structural mean is a mixture of type-specific components:

$$\mu_{\text{policy},i} = \pi \cdot \mu_{\text{Truster},i} + (1-\pi) \cdot \mu_{\text{Skeptic},i}$$

For treated units ($T_i = 1$):

$$\mu_{\text{Truster},i} = (\alpha_S + \alpha_\Delta) + \tau_{\text{T}}(\text{Gap}_i,\ \eta_{\text{trait},i}) + \mathbf{X}_i'\boldsymbol{\beta}_{\text{ctrl}}$$

$$\mu_{\text{Skeptic},i} = \alpha_S + \tau_{\text{S}}(\text{Gap}_i,\ \eta_{\text{trait},i}) + \mathbf{X}_i'\boldsymbol{\beta}_{\text{ctrl}}$$

The CATE for each class:

$$\tau_S(\text{Gap},\ \eta_{\text{trait}}) = \beta_{S,1} + \beta_{S,2} \cdot \text{Gap} + \beta_{S,3} \cdot \eta_{\text{trait}} + \beta_{S,4} \cdot \underbrace{\text{Gap} \times \eta_{\text{trait}}}_{\text{key interaction (H5)}}$$

# Results

# Unconditional Average Treatment Effect

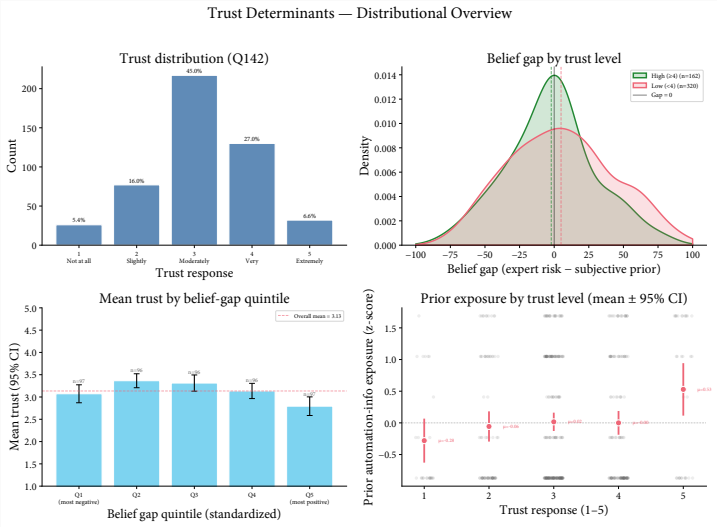
| Estimator | ATE ($d$) | SE/SD | 95% CI |
|---|---|---|---|
| OLS (no controls) | 0.27 | 0.07 | [0.14, 0.40] |
| OLS (with controls) | 0.24 | 0.06 | [0.13, 0.36] |
| SEM (no controls) | 0.26 | 0.07 | [0.13, 0.39] |
| SEM (with controls) | 0.24 | 0.06 | [0.13, 0.36] |
| BFMM ($\pi$-weighted) | 0.27 | 0.06 | [0.14, 0.39] |

# Unconditional Average Treatment Effect

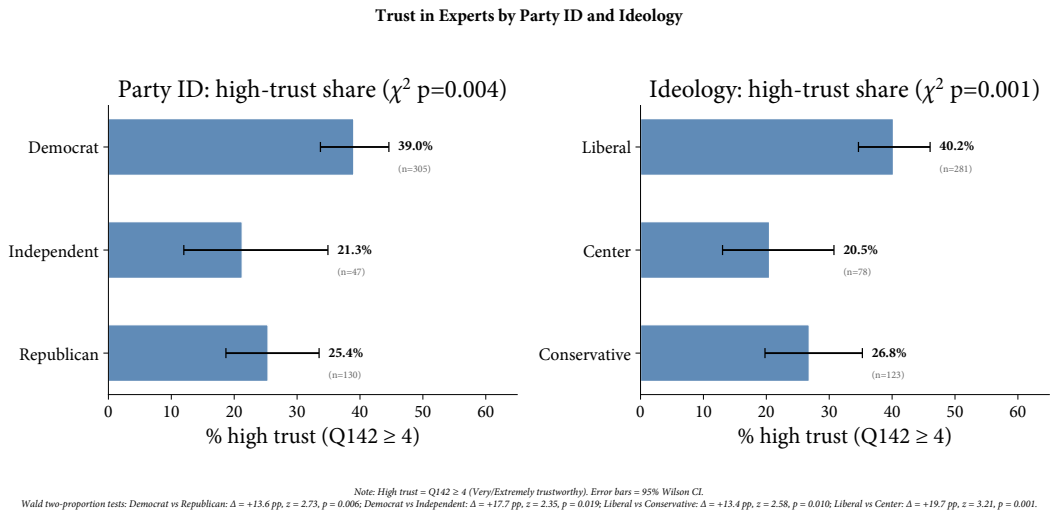
| Estimator | ATE ($d$) | SE/SD | 95% CI |
|---|---|---|---|
| OLS (no controls) | 0.27 | 0.07 | [0.14, 0.40] |
| OLS (with controls) | 0.24 | 0.06 | [0.13, 0.36] |
| SEM (no controls) | 0.26 | 0.07 | [0.13, 0.39] |
| SEM (with controls) | 0.24 | 0.06 | [0.13, 0.36] |
| BFMM ($\pi$-weighted) | 0.27 | 0.06 | [0.14, 0.39] |

All estimators converge at $d \approx 0.24$–$0.27$, indicating a positive but modest average treatment effect.

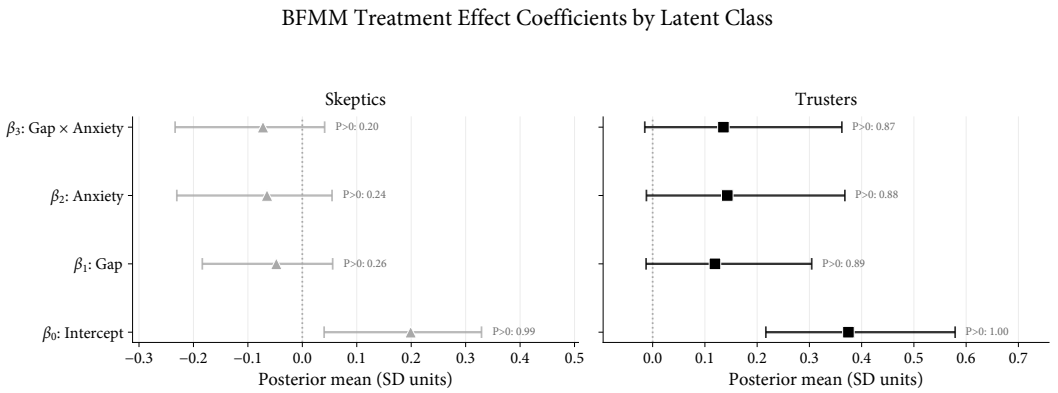
# First Stage: Latent Trust Classes



# Trust and Partisan Identity

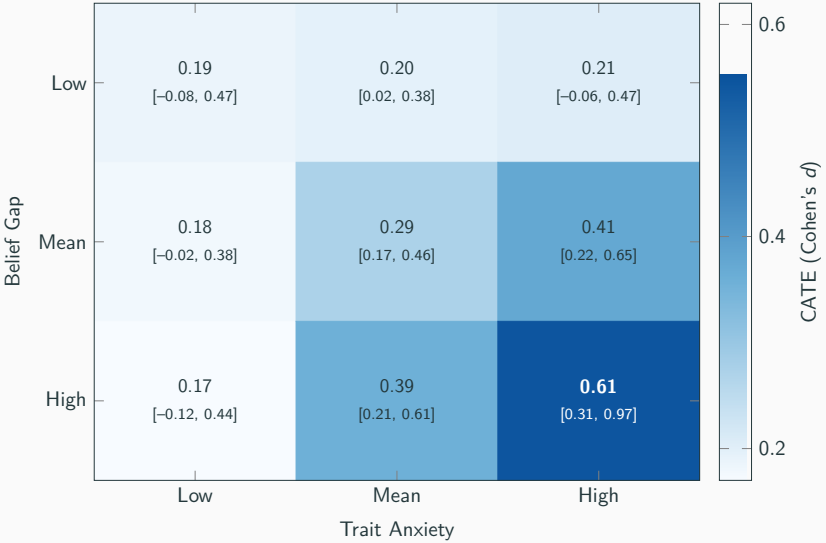


# BFMM Coefficients: Skeptics vs Trusters

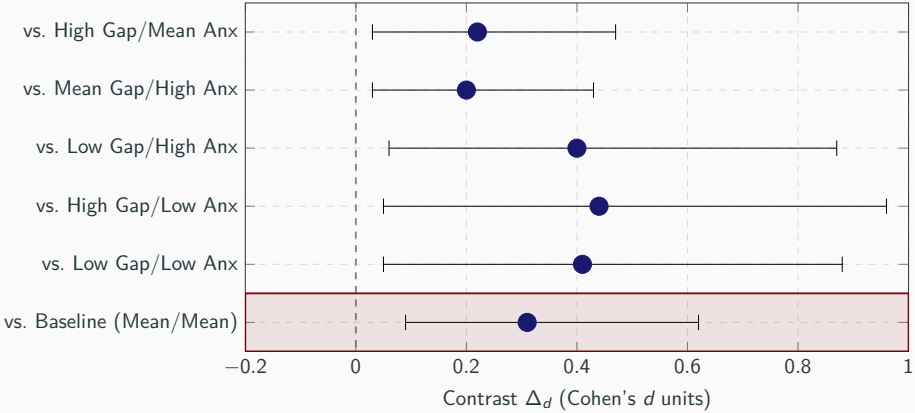



*Note: Dots = posterior mean. Bars = 90% credible interval. P>0 = posterior probability coefficient is positive.*

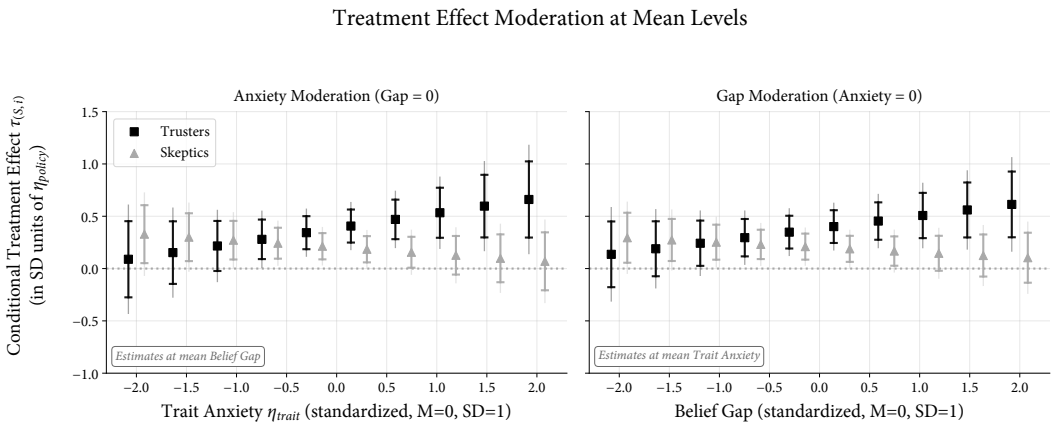
# CATE Surface: Treatment Effects Among Trusters



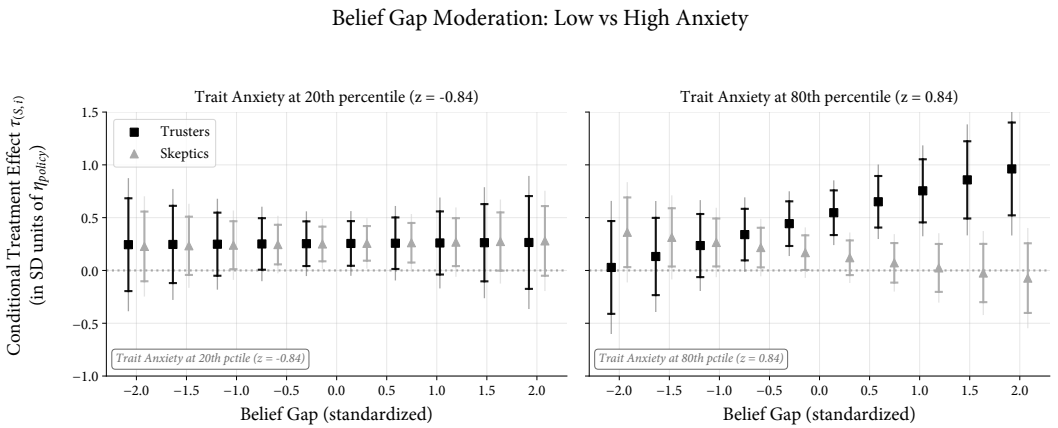
# CATE Contrasts: Maximum Activation Dominates



# Partial Effects: Gate 1 Is a Strict Filter

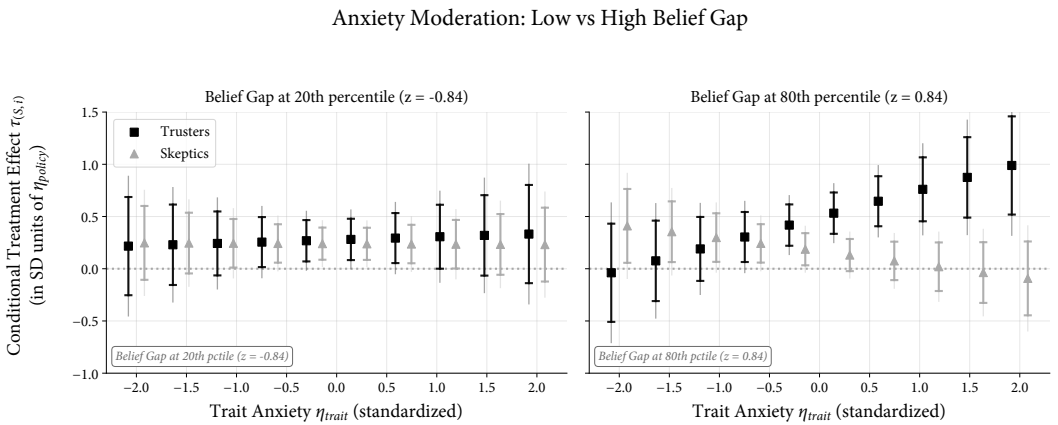


# Belief Gap Moderation: Low vs High Anxiety



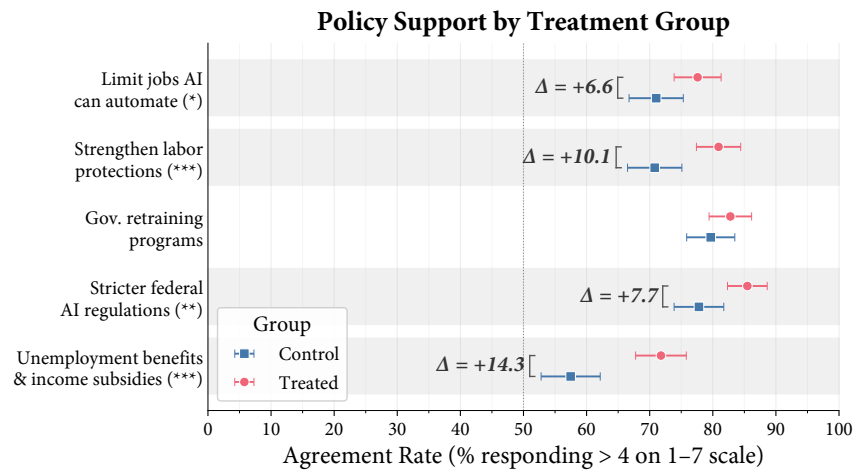

*Note: Points show posterior mean. Thick bars with caps = 83% CI. Thin lines = 95% CI. Dotted line = null effect.*

# Anxiety Moderation: Low vs High Belief Gap



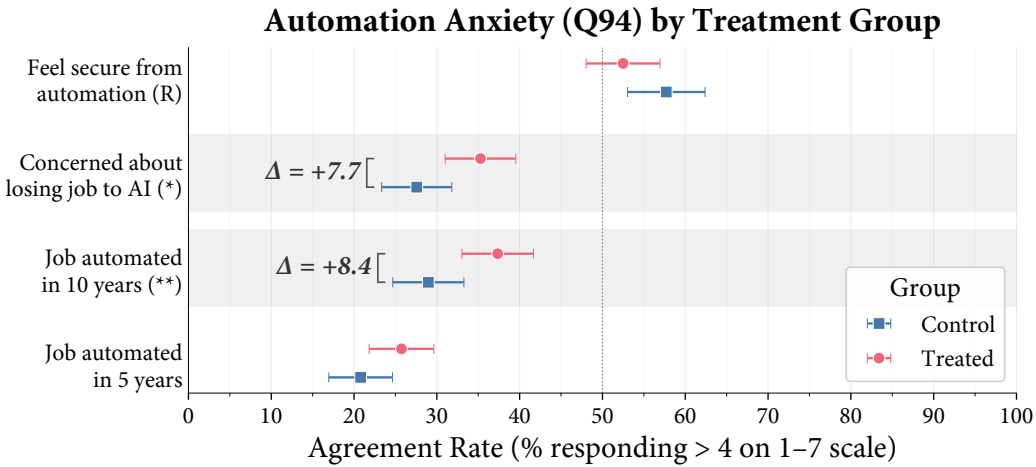

*Note: Points show posterior mean. Thick bars with caps = 83% CI. Thin lines = 95% CI. Dotted line = null effect.*

# Item-Level Effects: Policy Support (Q96)



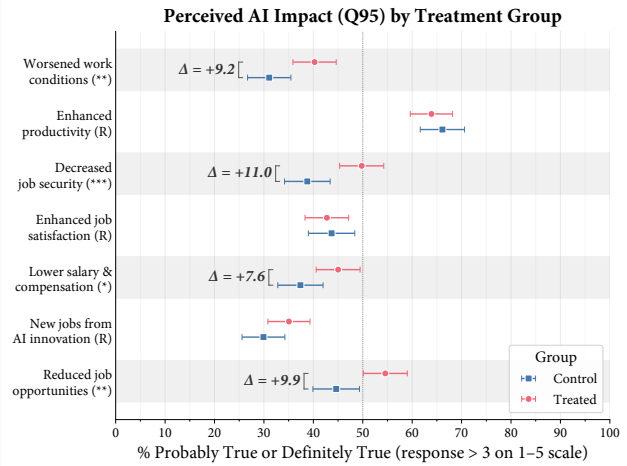

Note: * p < .05, ** p < .01, *** p < .001 (two-proportion z-test). Shaded rows = p < .05.
Δ = Treated − Control (pp). Error bars = 95% Wald CI.

# Item-Level Effects: Automation Anxiety (Q94)




Note: * p < .05, ** p < .01, *** p < .001 (two-proportion z-test). Shaded rows = p < .05.
Δ = Treated − Control (pp). Error bars = 95% Wald CI. (R) = reverse-worded item.

# Item-Level Effects: Perceived AI Impact (Q95)



# Discussion

# Three Pathways, Not Two

1. **Source rejection** (Skeptics, $\approx 62\%$ of sample)
   $\tau_S$ positive at intercept ($d \approx 0.20$) but *flat*: no moderation by Gap or Anxiety.
   Gate 1 blocks conditional activation.

# Three Pathways, Not Two

1. **Source rejection** (Skeptics, $\approx 62\%$ of sample)
   $\tau_S$ positive at intercept ($d \approx 0.20$) but *flat*: no moderation by Gap or Anxiety.
   Gate 1 blocks conditional activation.

2. **Shallow priming** (Trusters, low activation)
   $d \approx 0.18$–$0.21$ at low Gap or low Anxiety —
   information gets through but fails to mobilize demand.

# Three Pathways, Not Two

1. **Source rejection** (Skeptics, $\approx 62\%$ of sample)
   $\tau_S$ positive at intercept ($d \approx 0.20$) but *flat*: no moderation by Gap or Anxiety.
   Gate 1 blocks conditional activation.

2. **Shallow priming** (Trusters, low activation)
   $d \approx 0.18$–$0.21$ at low Gap or low Anxiety —
   information gets through but fails to mobilize demand.

3. **Deep persuasion** (Trusters, high activation)
   $d = 0.61$ [0.31, 0.97] at High Gap $\times$ High Anxiety —
   large informational shock meets dispositional vigilance.

# Three Pathways, Not Two

1. **Source rejection** (Skeptics, $\approx 62\%$ of sample)
   $\tau_S$ positive at intercept ($d \approx 0.20$) but *flat*: no moderation by Gap or Anxiety.
   Gate 1 blocks conditional activation.

2. **Shallow priming** (Trusters, low activation)
   $d \approx 0.18$–$0.21$ at low Gap or low Anxiety —
   information gets through but fails to mobilize demand.

3. **Deep persuasion** (Trusters, high activation)
   $d = 0.61$ [0.31, 0.97] at High Gap $\times$ High Anxiety —
   large informational shock meets dispositional vigilance.

The "null vs. works" framing in the literature is a false dichotomy.
All three pathways operate simultaneously within the same experiment.

# Limitations

- Single pre-registered experiment — no replication sample
- Online convenience sample (CloudResearch); liberal-leaning relative to U.S. population

# Limitations

- Single pre-registered experiment — no replication sample
- Online convenience sample (CloudResearch); liberal-leaning relative to U.S. population
- One-item trust measure ($Q_{142}$), post-treatment only $\Rightarrow$ collider risk (addressed structurally, not eliminated)
- Latent class prevalence $\pi$ has wide posterior: $[0.12, 0.75]$ — point estimate 0.38 but class sizes are uncertain

# Limitations

- Single pre-registered experiment — no replication sample
- Online convenience sample (CloudResearch); liberal-leaning relative to U.S. population
- One-item trust measure ($Q_{142}$), post-treatment only $\Rightarrow$ collider risk (addressed structurally, not eliminated)
- Latent class prevalence $\pi$ has wide posterior: $[0.12, 0.75]$ — point estimate 0.38 but class sizes are uncertain
- Untestable independence assumption: $S \perp \text{Gap}$ (dispositional trust does not cause prior beliefs)
- Policy battery skews toward protective/supportive items — may understate restrictive demand

# Takeaway

1. **The ATE masks conditional persuasion.**
   Unconditional $d \approx 0.27$ averages over an unmoderated subgroup (Skeptics, $d \approx 0.20$ flat) and a strongly moderated subgroup (Trusters, $d$ up to 0.61). The estimand matters.

# Takeaway

1. **The ATE masks conditional persuasion.**
   Unconditional $d \approx 0.27$ averages over an unmoderated subgroup (Skeptics, $d \approx 0.20$ flat) and a strongly moderated subgroup (Trusters, $d$ up to 0.61). The estimand matters.

2. **Trust operates as a discrete gate, not a continuous moderator.**
   Skeptics show a positive intercept ($d \approx 0.20$) but null moderation — the effect is *flat*, not conditional. Implications for any persuasion study with heterogeneous source acceptance.

# Takeaway

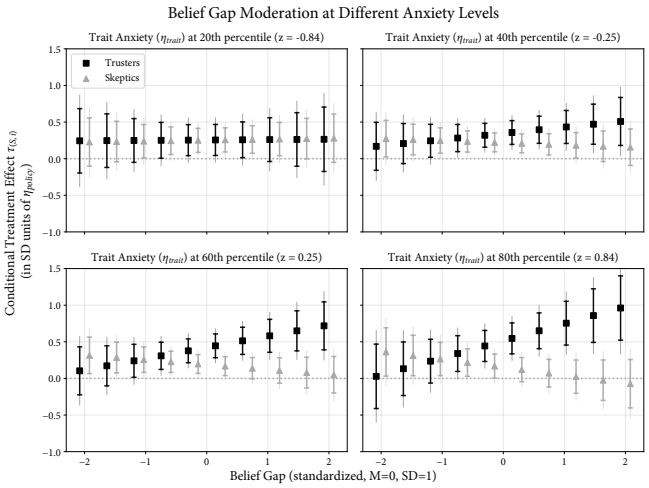
1. **The ATE masks conditional persuasion.**
   Unconditional $d \approx 0.27$ averages over an unmoderated subgroup (Skeptics, $d \approx 0.20$ flat) and a strongly moderated subgroup (Trusters, $d$ up to 0.61). The estimand matters.

2. **Trust operates as a discrete gate, not a continuous moderator.**
   Skeptics show a positive intercept ($d \approx 0.20$) but null moderation — the effect is *flat*, not conditional. Implications for any persuasion study with heterogeneous source acceptance.

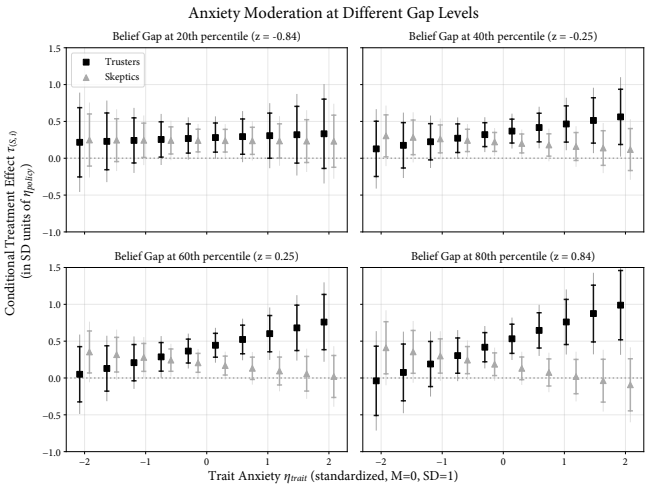
3. **The two-gate architecture may generalize.**
   Source credibility $\times$ information shock $\times$ dispositional anxiety is not automation-specific — same logic applies to climate, vaccines, trade. Open empirical question.

# Appendix

# Gap Slope Steepens With Anxiety



# Anxiety Slope Steepens With Belief Gap




*Note: Points show posterior mean. Thick bars with caps = 83% CI. Thin lines = 95% CI. Dotted line = null effect.*

# Instrument: Neutral Context Vignette (Step 1)

## Automation and Your Job

Artificial intelligence (AI) is getting better at doing work people used to do. Computers already answer customer questions, write basic code, and sort medical images. *As AI improves, some jobs may be partly-or fully-done by machines, while new kinds of jobs also appear.*

### How do experts judge a job's automation risk?

Experts look at the small tasks that make up a job. When those tasks are routine, clearly defined, and follow repeated patterns-and there are lots of examples for computers to learn from-AI systems can pick them up more easily.
The shorter and more repetitive a task is, the easier it is to automate.

| EXAMPLE JOB | TYPICAL TASK | AI RISK (TODAY) |
|---|---|---|
| IT Help Desk Technician | 10-minute help IT tickets that follow a script | *Higher Risk* – chatbots can handle most tickets |
| IT Systems Designer | Plan how the company's tech infrastructure will work over the next year | *Lower Risk* – needs long-term planning and team buy-in |

### Other factors also influence how likely a job is to be replaced by AI automation:

**Jobs are at greater risk when they:**

- Can be done entirely online, or
- Have lots of recorded data for AI to learn from.

**Jobs are less at risk when they:**

- Require in-person skills or dexterity, or
- Rely on human trust or empathy.

# Instrument: Expert Source Introduction (Step 3a)

**Now let's compare your view with what experts think.**

- Dr. Carl Benedikt Frey and Dr. Michael A. Osborne are researchers at the University of Oxford, where they investigate the impact of AI and automation on the labor market. *Their 2017 academic study, "The Future of Employment," is a landmark in this field—cited more than 17,000 times and widely used to gauge automation risk.*
- Here's how Dr. Frey and Dr. Osborne estimated job-automation risk:

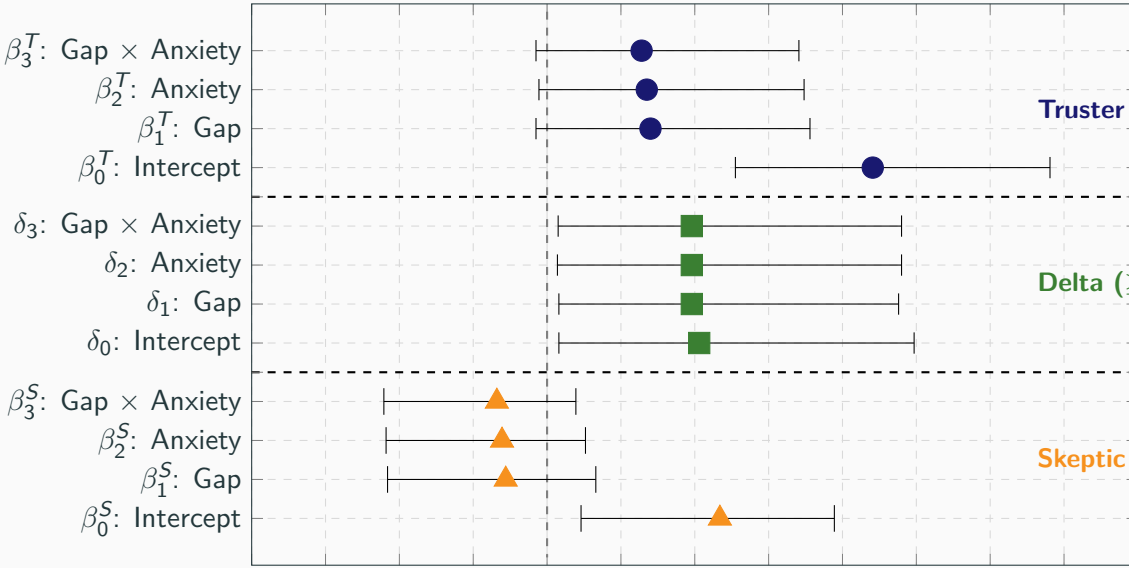
1. They started with the U.S. government's O*NET database, which lists the skills and day-to-day tasks for every job.
2. A panel of AI researchers first tagged 70 sample jobs as either "automatable" or "not yet automatable."
3. A computer model then learned the patterns: jobs heavy on routine, rule-based tasks, fine hand-eye work, or data processing score higher; jobs that rely on creative thinking, social insight, or physical dexterity score lower.

That model produced a score for each job—the higher the number, the sooner experts think machines could take over most of its tasks. *We use that score as the expert estimate for your occupation's relative risk of AI automation.*

**Your view vs. the expert estimate**

Below, you'll see where you placed your job on the automation-risk scale, compared to the expert estimate for your occupation.

# Model Coefficients (Forest Plot)



# BFMM Treatment Effects (Full Table)

| Parameter | Estimate | SD | 95% CI | $P(\beta > 0)$ |
|---|---|---|---|---|
| *Panel A: Skeptic Parameters ($\beta_j^S$)* | | | | |
| $\beta_0^S$: Intercept | 0.234 | 0.105 | [+0.047, +0.389] | 0.975 |
| $\beta_1^S$: Gap | -0.056 | 0.087 | [-0.216, +0.066] | 0.261 |
| $\beta_2^S$: Anxiety | -0.061 | 0.085 | [-0.218, +0.052] | 0.230 |
| $\beta_3^S$: Gap $\times$ Anxiety | -0.068 | 0.082 | [-0.221, +0.039] | 0.183 |
| *Panel B: Trust-Type Differential ($\beta_{\Delta,j} \geq 0$)* | | | | |
| $\beta_{\Delta,0}$: Intercept | 0.206 | 0.153 | [+0.016, +0.498] | 1.000 |
| $\beta_{\Delta,1}$: Gap | 0.196 | 0.147 | [+0.016, +0.476] | 1.000 |
| $\beta_{\Delta,2}$: Anxiety | 0.196 | 0.148 | [+0.015, +0.480] | 1.000 |
| $\beta_{\Delta,3}$: Gap $\times$ Anxiety | 0.196 | 0.148 | [+0.016, +0.481] | 1.000 |
| *Panel C: Truster Parameters ($\beta_j^T = \beta_j^S + \beta_{\Delta,j}$)* | | | | |
| $\beta_0^T$: Intercept | 0.441 | 0.131 | [+0.255, +0.680] | 1.000 |
| $\beta_1^T$: Gap | 0.140 | 0.117 | [-0.015, +0.356] | 0.924 |
| $\beta_2^T$: Anxiety | 0.135 | 0.113 | [-0.011, +0.348] | 0.929 |
| $\beta_3^T$: Gap $\times$ Anxiety | 0.129 | 0.112 | [-0.014, +0.341] | 0.919 |

# Partisan Heterogeneity

| Group | ATE (SD) | 90% CI | CATEhigh | P(>0.1) |
|---|---|---|---|---|
| *Panel A: Marginal Partisan Effects (ignoring trust class)* | | | | |
| Non-Democrat | 0.20 | [0.05, 0.34] | — | 87% |
| Democrat | 0.35 | [0.23, 0.48] | — | 100% |
| *Difference (Dem − Non-Dem)* | 0.15 | [-0.04, 0.34] | — | 68% |
| Non-Liberal | 0.21 | [0.07, 0.34] | — | 92% |
| Liberal | 0.35 | [0.22, 0.47] | — | 100% |
| *Difference (Lib − Non-Lib)* | 0.14 | [0.02, 0.29] | — | 67% |
| *Panel B: Cell-Specific Effects (Trust Class × Party)* | | | | |
| Skeptic × Non-Democrat | 0.14 | [-0.03, 0.30] | -0.21 [-0.56, 0.10] | 66% |
| Skeptic × Democrat | 0.29 | [0.12, 0.46] | 0.31 [-0.05, 0.63] | 97% |
| Truster × Non-Democrat | 0.32 | [0.12, 0.54] | 0.55 [0.12, 1.04] | 96% |
| Truster × Democrat | 0.47 | [0.17, 0.82] | 0.92 [0.28, 1.67] | 97% |
| *Panel C: Decomposition (Main Effects and Interaction)* | | | | |
| Trust Main (Truster − Skeptic) | 0.18 | [-0.07, 0.47] | — | 68% |
| Partisan Main (Dem − Non-Dem) | 0.15 | [-0.03, 0.34] | — | 68% |

# Mediation Analysis

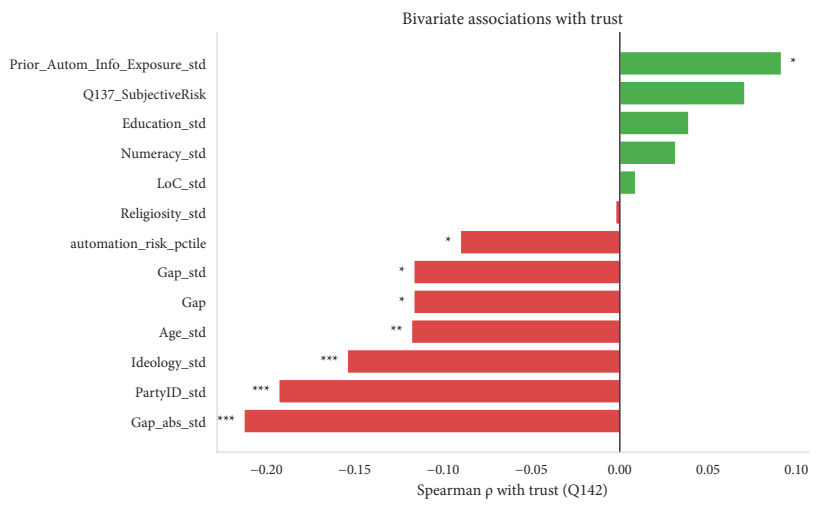
| Path | Coefficient | SE | $p$ | 90% CI | 95% CI |
|---|---|---|---|---|---|
| *Panel A: Treatment → Mediator (a paths)* | | | | | |
| Treatment → AutoAnxiety | 0.15* | 0.07 | .023 | [0.04, 0.26] | [0.02, 0.28] |
| Treatment → EgoThreat | 0.24*** | 0.06 | <.001 | [0.13, 0.34] | [0.11, 0.36] |
| Treatment → EgoOpp | −0.04 | 0.06 | .570 | [-0.14, 0.07] | [-0.16, 0.09] |
| *Panel B: Mediator → Policy Outcome (b paths, controlling for Treatment)* | | | | | |
| AutoAnxiety → Policy | 0.04 | 0.04 | .318 | [-0.03, 0.11] | [-0.04, 0.12] |
| EgoThreat → Policy | 0.11* | 0.05 | .011 | [0.04, 0.19] | [0.03, 0.20] |
| EgoOpp → Policy | −0.08* | 0.04 | .032 | [-0.14, -0.02] | [-0.15, -0.01] |
| *Panel C: Direct and Indirect Effects* | | | | | |
| Total Effect (c path) | 0.23*** | 0.06 | <.001 | [0.14, 0.33] | [0.12, 0.35] |
| Direct Effect (c' path) | 0.20*** | 0.06 | <.001 | [0.10, 0.29] | [0.08, 0.31] |
| *Indirect Effects (a × b):* | | | | | |
| via AutoAnxiety | 0.006 | — | — | — | — |
| via EgoThreat | 0.027 | — | — | — | — |

# Scale Reliability

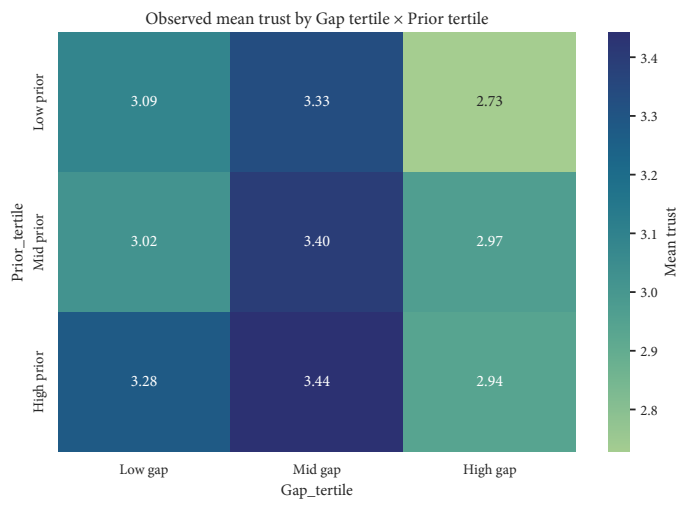
| Scale | $k$ | $\alpha$ | $\omega$ | CFI | TLI | RMSEA | SRMR | Verdict |
|---|---|---|---|---|---|---|---|---|
| Policy Demand (Q96) | 5 | 0.824 | 0.815 | 0.983 | 0.943 | 0.102 | 0.027 | good fit |
| Trait Anxiety (Q23) | 4 | 0.914 | 0.911 | 0.991 | 0.944 | 0.163 | 0.014 | good fit |
| Automation Anxiety (Q94) | 4 | 0.931 | 0.921 | 0.997 | 0.980 | 0.110 | 0.006 | good fit |
| Automation Impact (Q95) — 2-factor | 7 | — | — | 0.988 | 0.974 | 0.070 | 0.027 | good fit |
| Locus of Control (Q81) | 4 | 0.475 | 0.441 | 0.896 | 0.375 | 0.182 | 0.042 | poor fit |
| Economic Anxiety (Q24/Q25/Q26) | 3 | 0.705 | 0.727 | 1.000 | — | — | 0.000 | good fit |
| *Joint Measurement Model (6 factors, LoC excluded)* | | | | | | | | |
| 23 items, 6 factors | 23 | — | — | 0.958 | 0.949 | 0.054 | 0.048 | good fit |

*Notes:* $\alpha$ = Cronbach's alpha; $\omega$ = McDonald's omega (from CFA standardised loadings); CFI = Comparative Fit Index; TLI = Tucker-Lewis Index; RMSEA = Root Mean Square Error of Approximation; SRMR = Standardised Root Mean Square Residual. All CFA models include a priori correlated residuals for item pairs sharing a content sub-theme (see text). Verdict is based on a majority vote among CFI, TLI, and SRMR; $\chi^2$/df and RMSEA are excluded from or down-weighted in the verdict because they are inflated at $N > 300$ (Hu & Bentler 1999; Kline 2016). Thresholds: CFI, TLI $\geq 0.95$ (good), $\geq 0.90$ (acceptable); SRMR $\leq 0.08$ (good), $\leq 0.10$ (acceptable). LoC is excluded

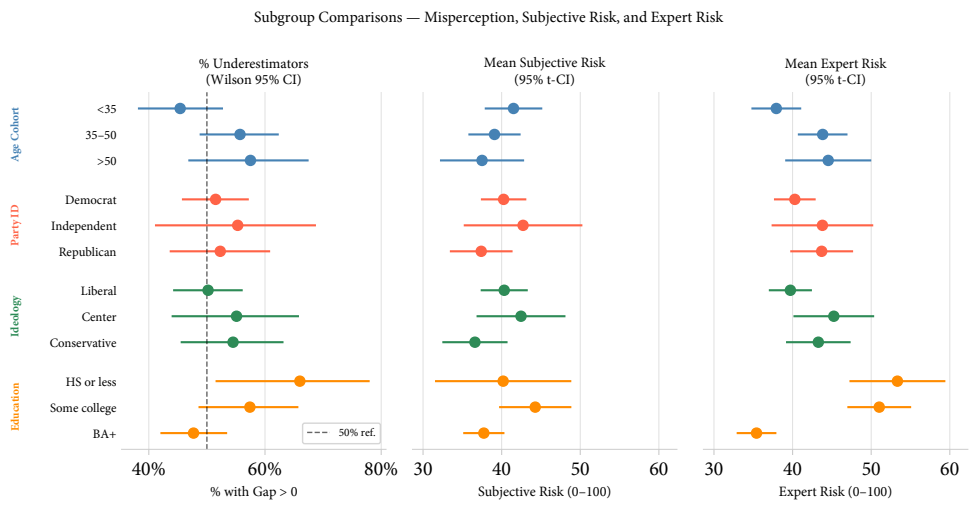
# Trust Correlates



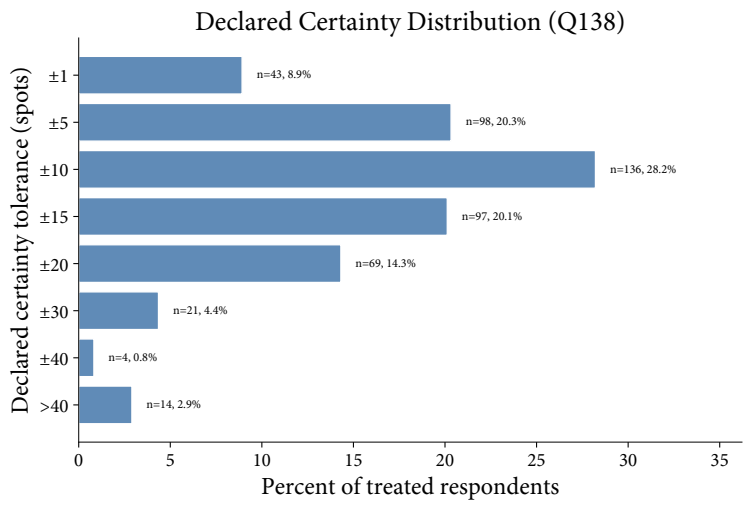
# Trust by Gap × Prior Exposure (Raw Means)



# Belief Gap Across Demographic Subgroups



# Declared Certainty Distribution (Q138)



# Prior Distributions

| Component | Parameter | Prior | Rationale |
|---|---|---|---|
| Trust Model | $\pi$ | $\text{Beta}(2, 2)$ | Weakly informative, centered at 0.5 |
| | $\delta$ | $\mathcal{N}^+(3, 3)$ | Positive separation; expects moderate effect |
| | $\boldsymbol{\kappa}$ | $\mathcal{N}(0, 1.5)$ | Regularized cutpoints |
| | $\boldsymbol{\gamma}_{\text{trust}}$ | $\mathcal{N}(0, 1)$ | Regularized covariate effects |
| Trait CFA | $\boldsymbol{\nu}_{\text{trait}}$ | $\mathcal{N}(3, 1.5)$ | Item intercepts (raw 1–7 scale) |
| | $\lambda_{\text{trait},1}$ | $= 1$ (fixed) | Reference indicator |
| | $\boldsymbol{\lambda}_{\text{trait},2:J}$ | $\mathcal{N}^+(0.9, 0.3)$ | Positive loadings near unity |
| | $\eta_{\text{trait},i}$ | $\mathcal{N}(0, 1)$ | Soft prior (scale stabilization) |
| Policy CFA | $\boldsymbol{\alpha}_{\text{policy}}$ | $\mathcal{N}(4, 2)$ | Item intercepts (raw 1–7 scale) |
| | $\lambda_{\text{policy},1}$ | $= 1$ (fixed) | Reference indicator |
| | $\boldsymbol{\lambda}_{\text{policy},2:J}$ | $\mathcal{N}^+(0.8, 0.3)$ | Positive loadings, moderate values |
| Structural | $\alpha_{\text{skeptic}}$ | $\mathcal{N}(0, 1)$ | Regularized baseline |
| | $\alpha_\Delta$ | $\mathcal{N}(0, 0.5)$, $\alpha_\Delta \geq -0.1$ | Tight prior; allows near-zero separation |
| | $\boldsymbol{\beta}^{(S)}$ | $\mathcal{N}(0, 0.5)$ | Regularized for realistic effect sizes |

# Sample Demographics

| Variable | Category | *n* | % |
|---|---|---|---|
| *Gender* | | | |
| | Male | 425 | 46.7 |
| | Female | 469 | 51.5 |
| | Non-binary / Other | 11 | 1.2 |
| | Prefer not to say | 5 | 0.5 |
| *Race / Ethnicity* | | | |
| | White | 629 | 69.1 |
| | Black or African American | 83 | 9.1 |
| | Hispanic or Latino | 1 | 0.1 |
| | Asian | 72 | 7.9 |
| | Native Hawaiian / Pacific Islander | 4 | 0.4 |
| | Other | 3 | 0.3 |
| | Prefer not to say | 51 | 5.6 |
| | Multiracial | 67 | 7.4 |
| *Education* | | | |
| | Less than high school | 4 | 0.4 |

# Descriptive Statistics

| Variable | *N* | Mean | SD | Min | Median | Max |
|---|---|---|---|---|---|---|
| *Panel A: Outcome and Treatment* | | | | | | |
| Policy Demand (mean, 1–5) | 910 | 5.49 | 1.15 | 1.00 | 5.60 | 7.00 |
| Policy Demand (std.) | 910 | -0.00 | 1.00 | -3.91 | 0.09 | 1.31 |
| Treatment Assignment | 910 | 0.53 | 0.50 | 0.00 | 1.00 | 1.00 |
| *Panel B: Psychological Scales* | | | | | | |
| Trait Anxiety (mean, 1–7) | 910 | 3.65 | 1.60 | 1.00 | 3.75 | 7.00 |
| Automation Anxiety (mean, 1–7) | 910 | 3.29 | 1.67 | 1.00 | 3.00 | 7.00 |
| Automation Impact F1 (mean, 1–5) | 910 | 3.11 | 1.07 | 1.00 | 3.25 | 5.00 |
| Automation Impact F2 (mean, 1–5) | 910 | 3.19 | 0.94 | 1.00 | 3.33 | 5.00 |
| Economic Anxiety (index, 1–5) | 910 | 2.79 | 0.93 | 1.00 | 2.67 | 5.00 |
| Locus of Control (index, 1–5) | 910 | 3.44 | 0.70 | 1.00 | 3.50 | 5.00 |
| Numeracy (correct / 3) | 910 | 2.48 | 0.74 | 0.00 | 3.00 | 3.00 |
| *Panel C: Treatment-Specific Variables* | | | | | | |
| Trust in Expert Source (1–5) | 482 | 3.13 | 0.95 | 1.00 | 3.00 | 5.00 |
| Subjective Risk ( | | | | | | |
| Prior Uncertainty ($\sigma$) | 468 | 11.8 | 7.4 | 1.0 | 10.0 | 40.0 |

# Full Balance — Continuous Covariates

| | Treated ($n = 482$) | | Control ($n = 428$) | | | |
|---|---|---|---|---|---|---|
| Covariate | Mean | SD | Mean | SD | Δ | SMD |
| Age (years) | 40.16 | 10.79 | 38.79 | 10.43 | 1.37 | 0.13 |
| Ideology (1=very liberal, 7=very conservative) | 3.19 | 1.77 | 3.36 | 1.85 | -0.17 | -0.09 |
| Education (1=less than HS, 5=graduate degree) | 3.73 | 0.91 | 3.64 | 0.90 | 0.09 | 0.10 |
| Numeracy score (0–3) | 2.50 | 0.71 | 2.46 | 0.77 | 0.04 | 0.06 |
| Party ID (1=strong Dem, 7=strong Rep) | 3.10 | 2.09 | 3.19 | 2.15 | -0.09 | -0.04 |
| Locus of control — internal index | 3.43 | 0.71 | 3.44 | 0.70 | -0.01 | -0.01 |
| Trait anxiety (mean of 4 items) | 3.76 | 1.61 | 3.53 | 1.59 | 0.23* | 0.14 |
| Economic anxiety index (std.) | 0.03 | 1.03 | -0.05 | 0.97 | 0.08 | 0.08 |
| Expert automation risk score (0–100) | 42.29 | 30.69 | 40.81 | 30.72 | 1.49 | 0.05 |
| Female (0/1) | 0.52 | 0.50 | 0.51 | 0.50 | 0.00 | 0.01 |
| White (0/1) | 0.70 | 0.46 | 0.68 | 0.47 | 0.02 | 0.05 |
| Risk tolerance index | -0.05 | 0.98 | -0.01 | 1.01 | -0.04 | -0.04 |
| Patience index | 0.01 | 0.99 | 0.01 | 1.03 | -0.01 | -0.01 |

*Note*: Δ = treated minus control mean. SMD = standardized mean difference (Cohen's *d*, pooled

# Full Balance — Categorical Covariates

| | Treated | | Control | |
|---|---|---|---|---|
| Category | *n* | % | *n* | % |
| *Gender* [$\chi^2(3) = 0.65,\ V = 0.027$] | | | | |
| Male | 223 | 46.3 | 202 | 47.2 |
| Female | 249 | 51.7 | 220 | 51.4 |
| Other | 7 | 1.5 | 4 | 0.9 |
| Prefer not to say | 3 | 0.6 | 2 | 0.5 |
| *Religious attendance* [$\chi^2(5) = 1.48,\ V = 0.040$] | | | | |
| Every week | 39 | 8.1 | 37 | 8.6 |
| Almost every week | 18 | 3.7 | 19 | 4.4 |
| 1–2/month | 30 | 6.2 | 26 | 6.1 |
| Few/year | 83 | 17.2 | 79 | 18.5 |
| Never (religious) | 71 | 14.7 | 53 | 12.4 |
| Never (not religious) | 241 | 50.0 | 214 | 50.0 |
| *Prior info about automation risk* [$\chi^2(1) = 0.00,\ V = 0.002$] | | | | |
| No | 287 | 59.5 | 254 | 59.3 |

# Scale Inter-Correlations

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
|---|---|---|---|---|---|---|---|
| (1) Policy Demand | — | | | | | | |
| (2) Trait Anxiety | 0.16*** | — | | | | | |
| (3) Automation Anxiety | 0.17*** | 0.21*** | — | | | | |
| (4) Auto Impact F1 | 0.27*** | 0.23*** | 0.72*** | — | | | |
| (5) Auto Impact F2 | −0.23*** | −0.16*** | −0.12*** | −0.35*** | — | | |
| (6) Locus of Control | −0.13*** | −0.34*** | −0.22*** | −0.26*** | 0.19*** | — | |
| (7) Economic Anxiety | 0.25*** | 0.47*** | 0.38*** | 0.42*** | −0.26*** | −0.40*** | — |

*Notes:* Spearman rank correlations, listwise complete ($N = 910$). Scale scores: Policy Demand and Trait Anxiety are item means; Automation Anxiety is the mean of four items (item 4 reverse-coded); Automation Impact F1 and F2 are subscale means from the two-factor Q95 model; Locus of Control is the 4-item internal index (items 3–4 reverse-coded); Economic Anxiety is the mean of Q24, Q25, and reverse-coded Q26. Significance: $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$.

# Trust Mixture Covariates

| Covariate | Mean | 95% CI (low) | 95% CI (high) |
|---|---|---|---|
| Gap (linear) | -0.144 | -0.317 | 0.026 |
| Gap$^2$ (quadratic) | -0.391 | -0.553 | -0.243 |
| Prior exposure (linear) | 0.043 | -0.173 | 0.272 |
| Prior exposure$^2$ (quadratic) | 0.294 | 0.026 | 0.566 |
| Class separation $\delta$ | 2.085 | 0.304 | 3.947 |

*Note:* Coefficients are gamma parameters from the trust (Q142) measurement model in the BFMM, with Gap and prior exposure standardized as in the main text.

# What Predicts the Belief Gap?

| Predictor | $\hat{\beta}$ | SE (HC3) | $p$ |
|---|---|---|---|
| Ideology (std.) | 0.415 | 1.747 | 0.812 |
| Education (std.) | -6.383*** | 1.539 | 0.000 |
| Age (std.) | 3.823* | 1.717 | 0.026 |
| Female | 1.665 | 3.408 | 0.625 |
| White | 0.933 | 3.433 | 0.786 |
| Numeracy (std.) | 2.482 | 2.214 | 0.263 |
| Economic anxiety (std.) | -2.528 | 1.936 | 0.192 |
| Trait anxiety (std.) | -1.030 | 2.009 | 0.608 |
| Locus of control (std.) | 3.274 | 1.883 | 0.083 |
| Prior automation info (std.) | -7.367*** | 1.598 | 0.000 |
| Constant | 0.666 | 3.191 | 0.835 |

*Note*: OLS regression, $n = 482$. $R^2 = 0.129$, adjusted $R^2 = 0.110$. $F(10, 471) = 6.98$, $p = 0.000$. SE = heteroskedasticity-robust (HC3). Dependent variable: belief gap (expert score − respondent subjective score). All continuous predictors standardized. Stars: $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$.

# Full Manipulation Checks

**Panel A. Q141 Response Distribution and Perception–Gap Alignment**

| Response category | $n$ | % |
|---|---|---|
| Significantly lower risk | 96 | 20.2 |
| Slightly lower risk | 72 | 15.2 |
| Similar risk | 70 | 14.7 |
| Slightly higher risk | 97 | 20.4 |
| Significantly higher risk | 140 | 29.5 |
| Spearman $r$(Q141, Gap) $= 0.737$, $p < 0.001$ | | |
| Direction agreement (excluding "similar"): 90.1% | | |

Treated sample only ($N = 475$).

# Full Manipulation Checks (Panels B–C)

**Panel B. Mean Q141 by Objective Gap Tercile**

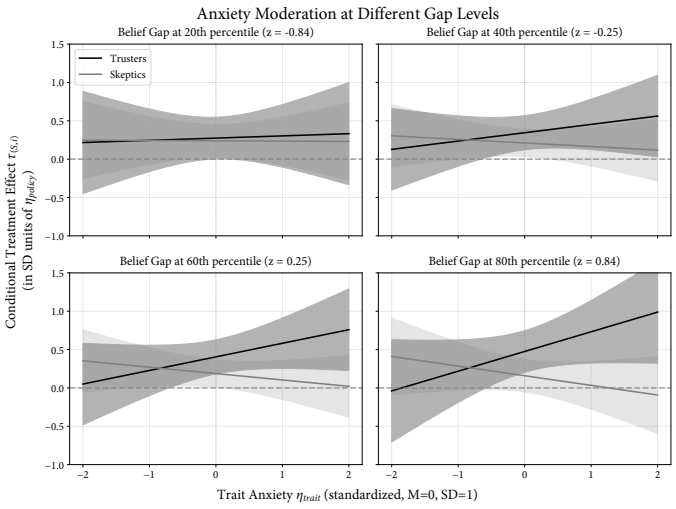
| Gap tercile | Mean | SD | *n* |
|---|---|---|---|
| Expert < Subjective | 1.92 | 1.29 | 158 |
| Mid Gap | 3.18 | 1.00 | 158 |
| Expert > Subjective | 4.61 | 0.76 | 159 |

**Panel C. Objective vs. Perceived Direction of the Gap**

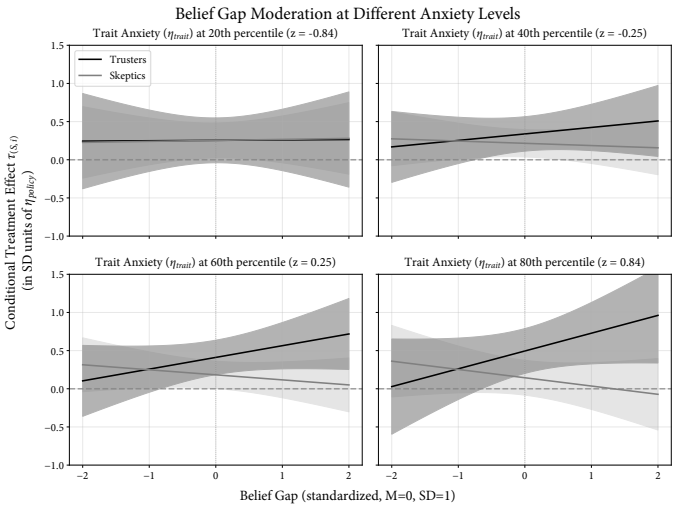
| | Perceived Higher | Perceived Lower | Similar | Total |
|---|---|---|---|---|
| Objective Higher | 169 | 5 | 11 | 185 |
| Objective Lower | 28 | 131 | 14 | 173 |
| Objective Similar | 40 | 32 | 45 | 117 |
| Total | 237 | 168 | 70 | 475 |

Q141 reverse-coded so higher values indicate "experts say higher risk."

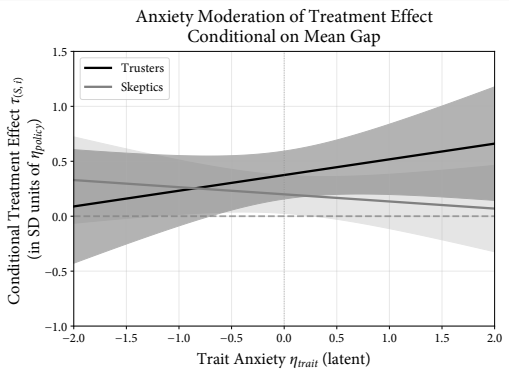
# Anxiety Partial Effects (Continuous Ribbons)

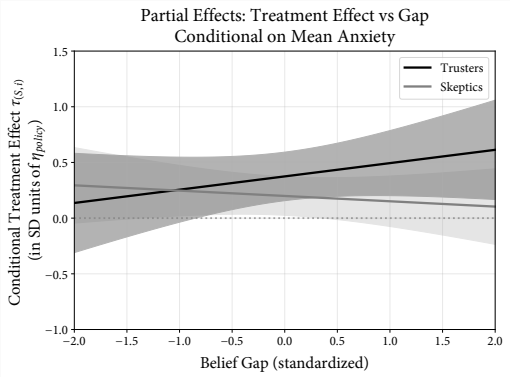


# Gap Partial Effects (Continuous Ribbons)

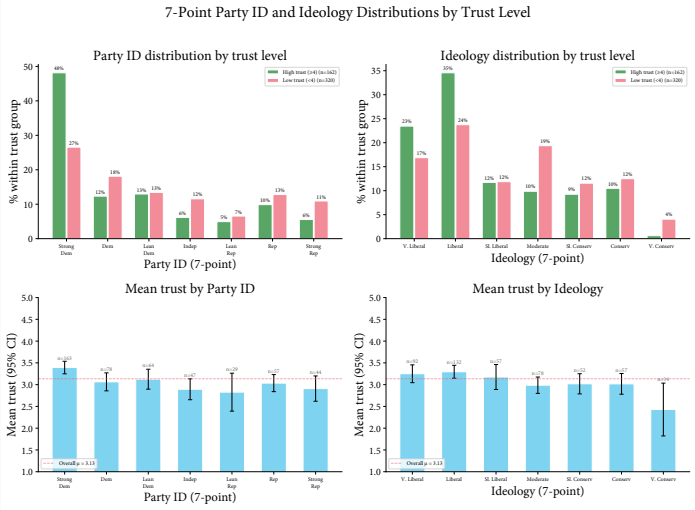


# Single-Moderator Partial Effects



# Trust by 7-Point Party ID and Ideology



# Automation Anxiety Item Effects

| Item | Control | Treatment | $\beta$ | SE | $p$ | 90% CI |
|---|---|---|---|---|---|---|
| Q94_1: Job automated in 5 years | 2.84 | 3.00 | 0.08 | 0.07 | .205 | [-0.02, 0.19] |
| Q94_2: Job automated in 10 years | 3.26 | 3.59 | 0.16* | 0.07 | .015 | [0.05, 0.27] |
| Q94_3: Concerned about losing job | 3.03 | 3.41 | 0.17** | 0.06 | .007 | [0.07, 0.28] |
| Q94_4: Feel secure (reversed) | 3.44 | 3.69 | 0.12† | 0.07 | .077 | [0.01, 0.22] |

*Notes:* Structural Equation Model with correlated residuals within scale and full covariate vector (ideology, party ID, age, gender, race, education, religiosity, locus of control, numeracy, trait anxiety). Items measured on 1–7 scale (Strongly Disagree to Strongly Agree). Control and Treatment columns show group means. $\beta$ is the standardized treatment effect. $N = 910$.

Q94_4 ("I feel secure that my occupation is safe from automation") is reverse-coded; higher values indicate *less* security.

† $p < .10$; * $p < .05$; ** $p < .01$; *** $p < .001$.

# Perceived AI Impact Item Effects

| Item | Control | Treatment | $\beta$ | SE | $p$ | 90% CI |
|---|---|---|---|---|---|---|
| *Threat Items (items 1, 3, 5, 7)* | | | | | | |
| Q95_1: Reduced job opportunities | 3.15 | 3.41 | 0.21** | 0.06 | .001 | [0.10, 0.31] |
| Q95_3: Lower salary | 2.95 | 3.18 | 0.20** | 0.06 | .002 | [0.09, 0.30] |
| Q95_5: Decreased job security | 2.96 | 3.27 | 0.23*** | 0.06 | .000 | [0.13, 0.34] |
| Q95_7: Worsened conditions | 2.82 | 3.08 | 0.21*** | 0.06 | .001 | [0.11, 0.32] |
| *Opportunity Items (items 2, 4, 6)* | | | | | | |
| Q95_2: New job opportunities | 2.88 | 2.88 | 0.03 | 0.06 | .651 | [-0.08, 0.14] |
| Q95_4: Enhanced satisfaction | 3.16 | 3.07 | -0.05 | 0.06 | .428 | [-0.16, 0.05] |
| Q95_6: Enhanced productivity | 3.61 | 3.54 | -0.07 | 0.07 | .271 | [-0.18, 0.04] |

*Notes:* Structural Equation Model with correlated residuals within subscale and full covariate vector (ideology, party ID, age, gender, race, education, religiosity, locus of control, numeracy, trait anxiety). Items measured on 1–5 scale. Control and Treatment columns show group means. $\beta$ is the standardized treatment effect. $N = 910$.

† $p < .10$; * $p < .05$; ** $p < .01$; *** $p < .001$.

# Full Mediation Path Decomposition

| Path | Coefficient | SE | $p$ | 90% CI | 95% CI |
|---|---|---|---|---|---|
| *Panel A: Treatment → Mediator (a paths)* | | | | | |
| Treatment → AutoAnxiety | 0.15* | 0.07 | .023 | [0.04, 0.26] | [0.02, 0.28] |
| Treatment → EgoThreat | 0.24*** | 0.06 | <.001 | [0.13, 0.34] | [0.11, 0.36] |
| Treatment → EgoOpp | −0.04 | 0.06 | .570 | [-0.14, 0.07] | [-0.16, 0.09] |
| *Panel B: Mediator → Policy Outcome (b paths, controlling for Treatment)* | | | | | |
| AutoAnxiety → Policy | 0.04 | 0.04 | .318 | [-0.03, 0.11] | [-0.04, 0.12] |
| EgoThreat → Policy | 0.11* | 0.05 | .011 | [0.04, 0.19] | [0.03, 0.20] |
| EgoOpp → Policy | −0.08* | 0.04 | .032 | [-0.14, -0.02] | [-0.15, -0.01] |
| *Panel C: Direct and Indirect Effects* | | | | | |
| Total Effect (c path) | 0.23*** | 0.06 | <.001 | [0.14, 0.33] | [0.12, 0.35] |
| Direct Effect (c' path) | 0.20*** | 0.06 | <.001 | [0.10, 0.29] | [0.08, 0.31] |
| *Indirect Effects (a × b):* | | | | | |
| via AutoAnxiety | 0.006 | — | — | — | — |
| via EgoThreat | 0.027 | — | — | — | — |

# Treatment Effects on Mechanism Scales

| Outcome | $\beta$ | SE | $p$ | 90% CI | 95% CI |
|---|---|---|---|---|---|
| *Panel A: Mechanism Scales* | | | | | |
| Automation Anxiety (Q94) | 0.167* | 0.066 | .012 | [0.06, 0.28] | [0.04, 0.30] |
| Egotropic Threat (Q95) | 0.248*** | 0.066 | $<$.001 | [0.14, 0.36] | [0.12, 0.38] |
| Egotropic Opportunity (Q95) | $-$0.054 | 0.066 | .418 | [-0.16, 0.06] | [-0.18, 0.08] |
| *Panel B: Primary Outcome (for comparison)* | | | | | |
| Policy Demand (Q96) | 0.275*** | 0.066 | $<$.001 | [0.17, 0.38] | [0.14, 0.40] |

*Notes:* OLS regression with HC3 robust standard errors. All outcomes standardized (mean $= 0$, SD $=$ 1). $\beta$ represents the treatment effect in standard deviation units.

Automation Anxiety: 4-item scale measuring job automation concerns (Q94_1–4, with Q94_4 reverse-coded). Egotropic Threat: 4-item scale measuring perceived negative AI career impacts (Q95_1, Q95_3, Q95_5, Q95_7). Egotropic Opportunity: 3-item scale measuring perceived positive AI career impacts (Q95_2, Q95_4, Q95_6).

* $p < .05$; ** $p < .01$; *** $p < .001$.

# Partisan Effects vs. Pre-Registered Criteria

| Effect | Est. | HDI $\neq 0$? | P(>0.05) | Decision |
|---|---|---|---|---|
| *Pass 1: Marginal Effects* | | | | |
| Democrat − Non-Democrat | +0.15 SD | Marginal | 82% | Suggestive |
| Liberal − Non-Liberal | +0.14 SD | Yes | 85% | Suggestive |
| *Pass 2–3: Interaction Model* | | | | |
| Trust Main Effect | +0.18 SD | No | 78% | Suggestive |
| Partisan Main Effect | +0.15 SD | No | 82% | Suggestive |
| Trust × Partisan | +0.01 SD | No | 41% | **Null** |
| *Within-Class Contrasts* | | | | |
| Within Skeptics | +0.15 SD | No | — | Suggestive |
| Within Trusters | +0.16 SD | No | — | Suggestive |

*Note:* "Suggestive" = consistent positive direction but fails preregistered criteria (90% HDI excludes zero *and* P(>0.05)> 0.95). "Null" = no directional evidence. P(>0.05) computed from posterior normal approximation.

# Partisan × Trust Heterogeneity

| Group | ATE (SD) | 90% CI | CATEhigh | P(>0.1) |
|---|---|---|---|---|
| *Panel A: Marginal Partisan Effects (ignoring trust class)* | | | | |
| Non-Democrat | 0.20 | [0.05, 0.34] | — | 87% |
| Democrat | 0.35 | [0.23, 0.48] | — | 100% |
| *Difference (Dem − Non-Dem)* | 0.15 | [-0.04, 0.34] | — | 68% |
| Non-Liberal | 0.21 | [0.07, 0.34] | — | 92% |
| Liberal | 0.35 | [0.22, 0.47] | — | 100% |
| *Difference (Lib − Non-Lib)* | 0.14 | [0.02, 0.29] | — | 67% |
| *Panel B: Cell-Specific Effects (Trust Class × Party)* | | | | |
| Skeptic × Non-Democrat | 0.14 | [-0.03, 0.30] | -0.21 [-0.56, 0.10] | 66% |
| Skeptic × Democrat | 0.29 | [0.12, 0.46] | 0.31 [-0.05, 0.63] | 97% |
| Truster × Non-Democrat | 0.32 | [0.12, 0.54] | 0.55 [0.12, 1.04] | 96% |
| Truster × Democrat | 0.47 | [0.17, 0.82] | 0.92 [0.28, 1.67] | 97% |
| *Panel C: Decomposition (Main Effects and Interaction)* | | | | |
| Trust Main (Truster − Skeptic) | 0.18 | [-0.07, 0.47] | — | 68% |
| Partisan Main (Dem − Non-Dem) | 0.15 | [-0.03, 0.34] | — | 68% |

**Temporary page!**

LaTeX was unable to guess the total number of pages correctly. As there was some unprocessed data that should have been added to the final page this extra page has been added to receive it.

If you rerun the document (without altering it) this surplus page will go away, becau LaTeX now knows how many pages to expect for this document.